

The minimum values for different datasets were obtained as given below:

U ₂ (V)	T (°C)	1	2	3	4	5	6	7	8
1	195	20.85	25.18	29.52	34.26	39.04	43.9	48.84	53.84
1	200	21.22	24.98	29.69	34.45	39.14	43.93	48.84	53.77
1	205	21.58	25.66	29.96	34.5	39.14	44.12	48.91	53.77
1.5	195	20.63	25.1	29.67	34.28	39.04	44	48.86	53.87
1.5	200	21.68	25.61	29.96	34.5	39.34	44	48.86	53.79
1.5	205	22.29	26.23	30.38	34.67	39.36	44.22	48.98	53.89

The minimum separations and their averages were tabulated as:

U ₂ (V)	T (°C)	1	2	3	4	5	6	7	Average of Separations
1	195	4.33	4.34	4.74	4.78	4.86	4.94	5	4.71
1	200	3.76	4.71	4.76	4.69	4.79	4.91	4.93	4.65
1	205	4.08	4.3	4.54	4.64	4.98	4.79	4.86	4.60
1.5	195	4.47	4.57	4.61	4.76	4.96	4.86	5.01	4.75
1.5	200	3.93	4.35	4.54	4.84	4.66	4.86	4.93	4.59
1.5	205	3.94	4.15	4.29	4.69	4.86	4.76	4.91	4.51

Average of all the separations = $(4.71 + 4.65 + 4.60 + 4.75 + 4.59 + 4.51)/6 = 4.64 \text{ V}$

The (N-1) Standard Deviation of the 6 average separations around 4.64V:

$$(((4.71-4.64)^2 + (4.65-4.64)^2 + (4.60-4.64)^2 + (4.75-4.64)^2 + (4.59-4.64)^2 + (4.51-4.64)^2)/5)^{(1/2)}$$

= 0.087 ~ 0.09

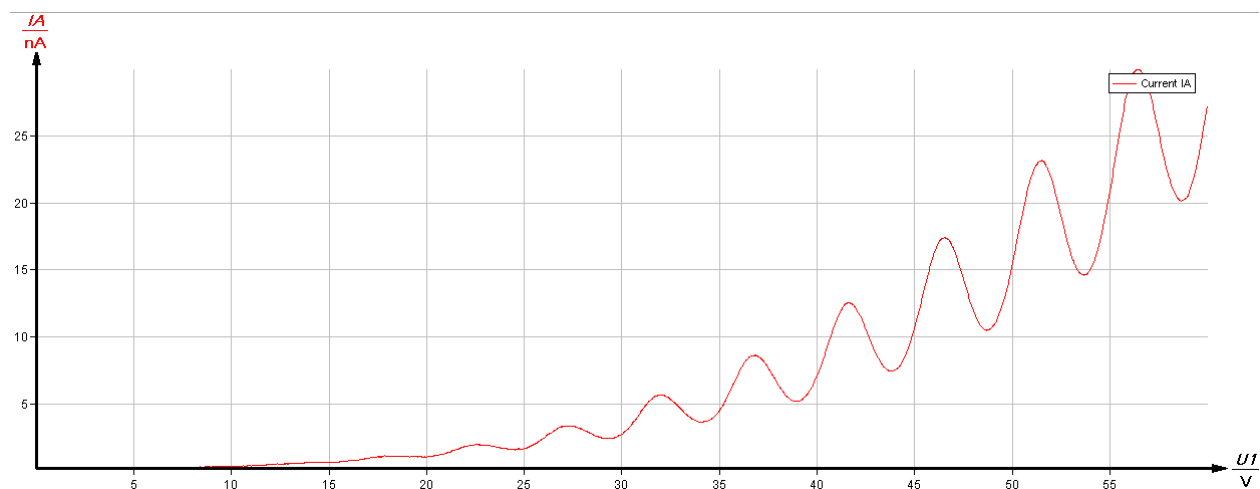
Therefore, the separation is found to be 4.64 ± 0.09

Plots for $U_2 = 1.0 \text{ V}$

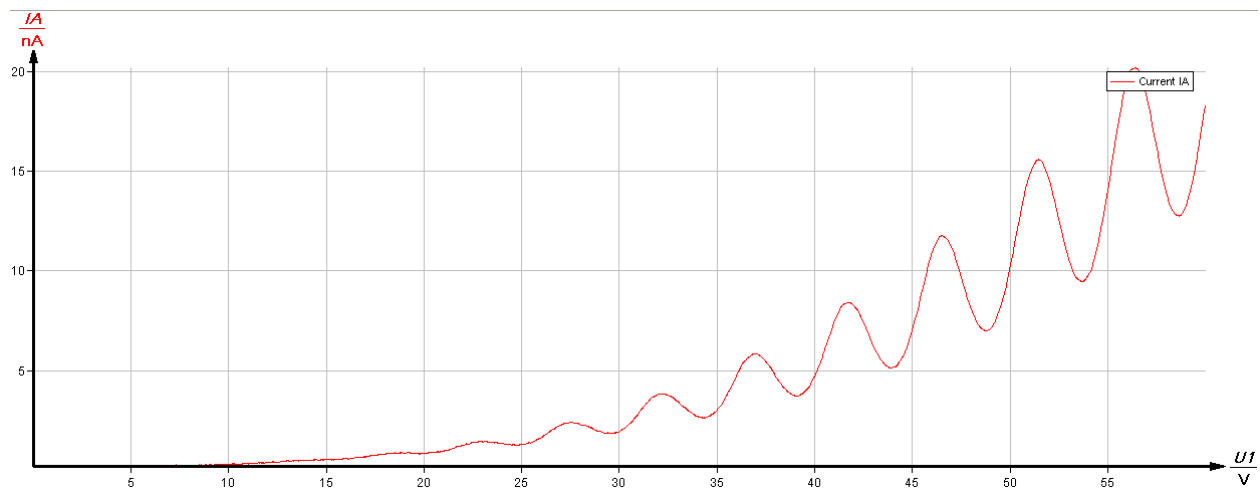
T = 195°C



T = 200°C

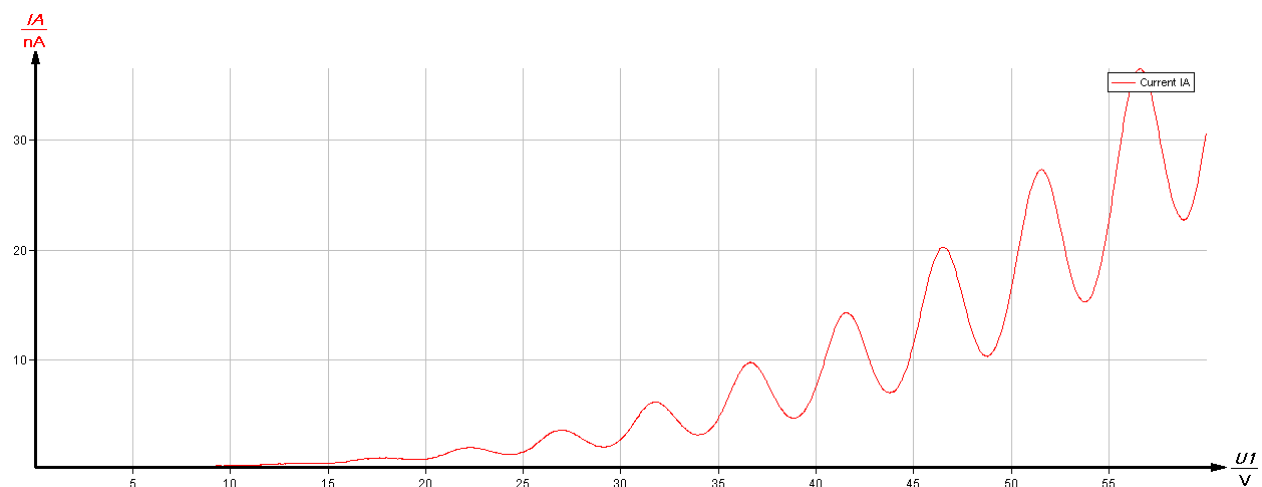


T = 205°C

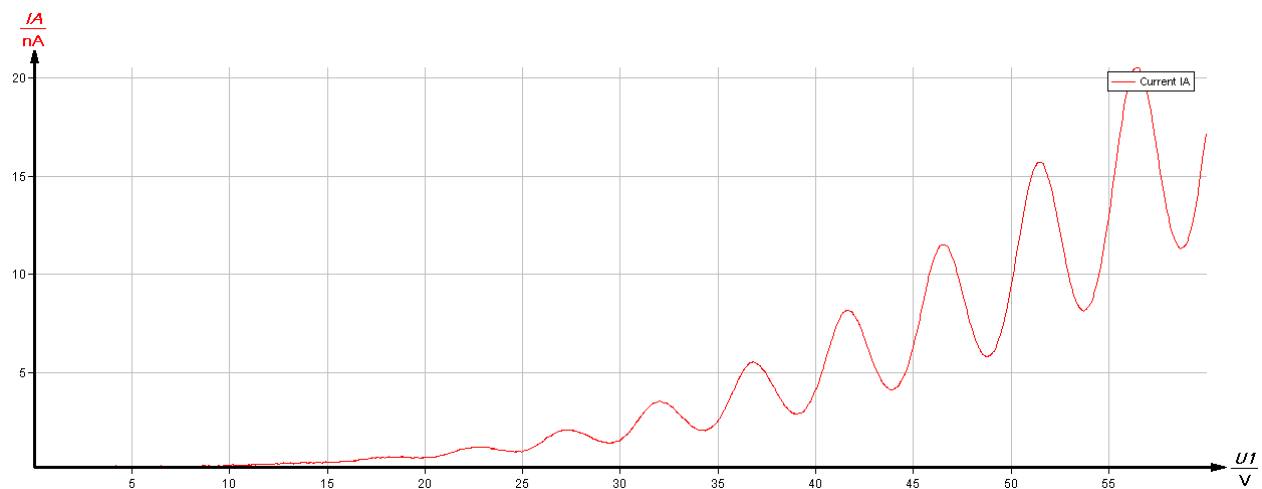


Plots for $U_2 = 1.5 \text{ V}$

T = 195°C



T = 200°C



xT = 205°C

